

Project 2

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1 Introduction

This report presents a regression and symbolic regression analysis across three benchmark datasets. The analysis includes multiple regression, regularized regression using Ridge and Lasso, forward feature selection, transformed regression, and symbolic regression. ScalaTion (Scala 3) is used across all five modeling approaches, while Python's `statsmodels` is used for multiple and regularized regression and `PySR` is used for symbolic regression. Across the required cases, model results and evaluation measures are presented and discussed to examine model fit, predictive performance, and the effects of different regression techniques.

2 Multiple Regression

Multiple regression is used to predict a single outcome from multiple independent variables while estimating the association between each predictor and the response conditional on the other predictors. For each dataset, a multiple regression model was fitted using all available predictors in both ScalaTion and `statsmodels`, giving six required regression models. Additional simple regression models were fitted using the two predictors with the largest absolute correlations with the response variable. The same predictor selections were used for the ScalaTion and `statsmodels` analyses. All R^2 values reported for the full regression models are in-sample values computed using the full datasets and are therefore not directly comparable to held-out test-set metrics reported in later sections.

2.1 ScalaTion

The ScalaTion implementation used an intercept term and all available predictors for each of the multiple regression models. The fitted models were evaluated using the reported R^2 , adjusted R^2 , residual standard error, and fitted coefficient statistics. Simple regression models were also fitted for the two predictors with the largest absolute correlations with each response variable.

2.1.1 Auto MPG

For the Auto MPG dataset, the full multiple regression model uses cylinders, displacement, horsepower, weight, acceleration, model year, and origin to predict MPG. The resulting model showed a strong overall fit, with an R^2 of approximately 0.821. Substantial multicollinearity was present among several vehicle characteristics. The VIF values from ScalaTion were above 10 for cylinders, displacement, and weight, with displacement having the highest VIF at approximately 21.84. The two predictors with the largest absolute correlations with MPG were weight and displacement. Both simple regression models showed a negative relationship with MPG, with higher vehicle weight and displacement corresponding to lower predicted MPG.

For Auto MPG, weight, model year, and origin were statistically significant predictors in the full model at $p < 0.001$. Weight had a negative coefficient, while model year and origin had positive coefficients. Displacement also had a positive coefficient and was significant at $p = 0.008$, despite its negative simple-regression relationship with MPG. This difference illustrates the effect of controlling for correlated predictors in a multiple regression model. The high VIF values for displacement, cylinders, and weight indicate substantial multicollinearity, so individual coefficient estimates should be interpreted with caution.

Table 1: Auto MPG full regression coefficients from ScalaTion.

Predictor	Estimate	Std. Error	t	p	VIF
Intercept	-17.2184	4.6443	-3.71	0.0002	—
Cylinders	-0.4934	0.3233	-1.53	0.1270	10.74
Displacement	0.01990	0.00752	2.65	0.0081	21.84
Horsepower	-0.01695	0.01379	-1.23	0.2189	9.94
Weight	-0.00647	0.00065	-9.93	0.001	10.83
Acceleration	0.0806	0.0988	0.82	0.4150	2.63
Model year	0.7508	0.0510	14.73	0.001	1.24
Origin	1.4261	0.2781	5.13	0.001	1.77

OLS Regression Results						
Dep. Variable:	mpg	R-squared:	0.693			
Model:	OLS	Adj. R-squared:	0.692			
Method:	Least Squares	F-statistic:	878.8			
Date:	Mon, 21 Sep 2026	Prob (F-statistic):	6.02e-102			
Time:	09:07:01	Log-Likelihood:	-1130.0			
No. Observations:	392	AIC:	2264.			
Df Residuals:	390	BIC:	2272.			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	46.2165	0.799	57.867	0.000	44.646	47.787
weight	-0.0076	0.000	-29.645	0.000	-0.008	-0.007
Omnibus:	41.682	Durbin-Watson:	0.000			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	60.039			
Skew:	0.727	Prob(JB):	9.18e-14			
Kurtosis:	4.251	Cond. No.	1.13e+04			

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.13e+04. This might indicate that there are strong multicollinearity or other numerical problems.

Figure 1: Auto MPG simple regression using vehicle weight.

2.1.2 Concrete

For the Concrete dataset, the full multiple regression model used cement, blast furnace slag, fly ash, water, superplasticizer, coarse aggregate, fine aggregate, and age to predict compressive strength. The resulting model showed an R^2 of approximately 0.615. The two predictors with the largest absolute correlations with compressive strength were cement and superplasticizer, and both showed positive simple regression relationships with compressive strength. The Concrete predictors also showed some multicollinearity. Several VIF values were above 5, including cement, blast furnace slag, fly ash, water, coarse aggregate, and fine aggregate. These elevated VIF values indicate that several predictors are correlated with the other predictors in the model, which can make individual coefficient estimates more difficult to interpret.

For Concrete, cement, blast furnace slag, fly ash, water, superplasticizer, and age were statistically significant predictors at the 0.05 level. Water had a negative coefficient, while the other significant predictors had positive coefficients. Coarse aggregate and fine aggregate were not significant at the 0.05 level, with p-values of approximately 0.055 and 0.060, respectively. The elevated VIF values for several material variables indicate substantial correlation among some of the predictors.

OLS Regression Results						
Dep. Variable:	mpg	R-squared:	0.648			
Model:	OLS	Adj. R-squared:	0.647			
Method:	Least Squares	F-statistic:	718.7			
Date:	Mon, 21 Sep 2026	Prob (F-statistic):	1.66e-90			
Time:	09:07:01	Log-Likelihood:	-1156.4			
No. Observations:	392	AIC:	2317.			
Df Residuals:	390	BIC:	2325.			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	35.1206	0.494	71.033	0.000	34.149	36.093
displacement	-0.0601	0.002	-26.008	0.000	-0.064	-0.056
Omnibus:	41.308	Durbin-Watson:	0.926			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	61.139			
Skew:	0.709	Prob(JB):	5.30e-14			
Kurtosis:	4.317	Cond. No.	466.			

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Figure 2: Auto MPG simple regression using vehicle displacement.

Table 2: Concrete full regression coefficients from ScalaTion.

Predictor	Estimate	Std. Error	t	p	VIF
Intercept	-23.1638	26.5884	-0.87	0.3836	—
Cement	0.1198	0.0085	14.11	0.001	7.49
Blast furnace slag	0.1038	0.0101	10.25	0.001	7.28
Fly ash	0.0879	0.0126	6.99	0.001	6.17
Water	-0.1503	0.0402	-3.74	0.0002	7.00
Superplasticizer	0.2907	0.0935	3.11	0.0019	2.97
Coarse aggregate	0.0180	0.0094	1.92	0.0549	5.08
Fine aggregate	0.0202	0.0107	1.88	0.0597	7.01
Age	0.1142	0.0054	21.05	0.001	1.12

```

=====
                        OLS Regression Results
=====
Dep. Variable:      compressive_strength_mpa      R-squared:      0.248
Model:              OLS                          Adj. R-squared:  0.247
Method:             Least Squares                F-statistic:    338.7
Date:               Mon, 21 Sep 2026              Prob (F-statistic): 1.32e-65
Time:               09:07:02                      Log-Likelihood: -4214.6
No. Observations:   1030                          AIC:           8433.
Df Residuals:       1028                          BIC:           8443.
Df Model:            1
Covariance Type:    nonrobust

=====
              coef      std err      t      P>|t|      [0.025      0.975]
-----
const      13.4428      1.297      10.365      0.000      10.898      15.988
cement      0.0796      0.004      18.405      0.000      0.071      0.088
=====
Omnibus:      19.695      Durbin-Watson:      1.012
Prob(Omnibus):      0.000      Jarque-Bera (JB):      17.890
Skew:          0.271      Prob(JB):      0.000130
Kurtosis:      2.649      Cond. No.      861.
=====

```

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Figure 3: Concrete simple regression using cement.

2.1.3 Airfoil

For the Airfoil dataset, the full multiple regression model used frequency, angle of attack, chord length, free-stream velocity, and suction displacement thickness to predict scaled sound pressure. The full model produced an R^2 of approximately 0.516. The two predictors with the largest absolute correlations with scaled sound pressure were frequency and suction displacement thickness. Both simple regression models showed a negative relationship with the response. Compared with the other two datasets, the Airfoil predictors showed lower multicollinearity, with all VIF values below 3.5. Angle of attack had the highest VIF at approximately 3.44, while the remaining predictors had VIF values below 2.6.

For Airfoil, all five predictors were statistically significant at $p < 0.001$. Frequency, angle of attack, chord length, and suction displacement thickness had negative coefficients, while free-stream velocity had a positive coefficient. Despite the statistical significance of the predictors, the model explained approximately 51.6% of the variation in scaled sound pressure. The relatively modest VIF values suggest that multicollinearity was less severe than in the Auto MPG and Concrete models.

```

=====
                        OLS Regression Results
=====
Dep. Variable:      compressive_strength_mpa      R-squared:      0.134
Model:              OLS                          Adj. R-squared:  0.133
Method:             Least Squares                F-statistic:    159.1
Date:               Mon, 21 Sep 2026              Prob (F-statistic): 5.08e-34
Time:               09:07:02                      Log-Likelihood: -4287.1
No. Observations:   1030                          AIC:           8578.
Df Residuals:       1028                          BIC:           8588.
Df Model:            1
Covariance Type:    nonrobust

=====
              coef      std err      t      P>|t|      [0.025      0.975]
-----
const      29.4668      0.699      42.165      0.000      28.095      30.838
superplasticizer      1.0239      0.081      12.614      0.000      0.865      1.183
=====
Omnibus:      24.083      Durbin-Watson:      1.052
Prob(Omnibus):      0.000      Jarque-Bera (JB):      24.387
Skew:          0.353      Prob(JB):      5.06e-06
Kurtosis:      2.738      Cond. No.      12.5
=====

```

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Figure 4: Concrete simple regression using superplasticizer.

Table 3: Airfoil full regression coefficients from ScalaTion.

Predictor	Estimate	Std. Error	t	p	VIF
Intercept	132.8338	0.5447	243.87	0.001	—
Frequency	-0.00128	0.00004	-30.45	0.001	1.14
Angle of attack	-0.4219	0.0389	-10.85	0.001	3.44
Chord length	-35.6880	1.6304	-21.89	0.001	1.51
Free-stream velocity	0.0999	0.0081	12.28	0.001	1.04
Suction displacement thickness	-147.3005	15.0147	-9.81	0.001	2.53

```

=====
OLS Regression Results
=====
Dep. Variable:    scaled_sound_pressure_db    R-squared:        0.153
Model:            OLS                        Adj. R-squared:    0.152
Method:            Least Squares              F-statistic:       270.4
Date:              Mon, 21 Sep 2026            Prob (F-statistic): 5.36e-56
Time:              09:07:02                    Log-Likelihood:    -4910.5
No. Observations: 1503                        AIC:               9825.
Df Residuals:      1501                       BIC:               9836.
Df Model:           1
Covariance Type:   nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
const	127.3037	0.222	572.944	0.000	126.868	127.740
frequency_hz	-0.0009	5.2e-05	-16.444	0.000	-0.001	-0.001

```

=====
Omnibus:            4.973    Durbin-Watson:       0.255
Prob(Omnibus):      0.083    Jarque-Bera (JB):    5.002
Skew:               -0.126    Prob(JB):             0.020
Kurtosis:           2.873    Cond. No.             5.80e+03
=====
Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 5.8e+03. This might indicate that there are
strong multicollinearity or other numerical problems.

```

Figure 5: Airfoil simple regression using frequency.

2.2 statsmodels

The same three datasets and predictor selections were analyzed using Python’s `statsmodels` ordinary least squares (OLS) implementation. The full models used all available predictors, while the simple models used the two predictors with the largest absolute response correlations. The full Auto MPG model achieved an R^2 of 0.8215 and an adjusted R^2 of 0.8182, with a residual standard error of approximately 3.33 MPG. The Concrete model achieved an R^2 of 0.6155 and an adjusted R^2 of 0.6125, with a residual standard error of approximately 10.40 MPa. The Airfoil model achieved an R^2 of 0.5157 and an adjusted R^2 of 0.5141, with a residual standard error of approximately 4.81 dB.

The `statsmodels` simple regression results were consistent with the correlation analysis. For Auto MPG, weight produced an R^2 of 0.6926, while displacement produced an R^2 of 0.6482. For Concrete, cement produced an R^2 of 0.2478, while superplasticizer produced an R^2 of 0.1340. For Airfoil, frequency produced an R^2 of 0.1527, while suction displacement thickness produced an R^2 of 0.0978. All six simple regression models had statistically significant predictor coefficients with $p < 0.001$.

2.3 Results Comparison

The ScalaTion and `statsmodels` implementations produced consistent results across all three datasets. Both implementations produced the same full-model R^2 , adjusted R^2 , residual standard error, and F-statistic to

```

=====
OLS Regression Results
=====
Dep. Variable:    scaled_sound_pressure_db    R-squared:        0.098
Model:            OLS                        Adj. R-squared:    0.097
Method:            Least Squares              F-statistic:       162.6
Date:              Mon, 21 Sep 2026            Prob (F-statistic): 1.92e-35
Time:              09:07:02                    Log-Likelihood:    -4957.6
No. Observations: 1503                        AIC:               9919.
Df Residuals:      1501                       BIC:               9930.
Df Model:           1
Covariance Type:   nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
const	126.6632	0.222	571.527	0.000	126.228	127.098
suction_displacement_thickness_m	-164.0275	12.862	-12.753	0.000	-169.256	-138.798

```

=====
Omnibus:            26.070    Durbin-Watson:       0.364
Prob(Omnibus):      0.000    Jarque-Bera (JB):    25.468
Skew:               -0.280    Prob(JB):             2.95e-06
Kurtosis:           2.694    Cond. No.             76.1
=====
Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```

Figure 6: Airfoil simple regression using suction displacement thickness.

Table 4: Full multiple regression evaluation results from `statsmodels`.

Dataset	n	R^2	Adjusted R^2	Residual SE	F
Auto MPG	392	0.8215	0.8182	3.3277	252.43
Concrete	1030	0.6155	0.6125	10.3998	204.27
Airfoil	1503	0.5157	0.5141	4.8089	318.82

Table 5: Auto MPG full regression coefficients from `statsmodels`.

Predictor	Estimate	Std. Error	t	p
Intercept	-17.2184	4.644	-3.707	0.001
Cylinders	-0.4934	0.323	-1.526	0.128
Displacement	0.0199	0.008	2.647	0.008
Horsepower	-0.0170	0.014	-1.230	0.220
Weight	-0.0065	0.001	-9.929	0.001
Acceleration	0.0806	0.099	0.815	0.415
Model year	0.7508	0.051	14.729	0.001
Origin	1.4261	0.278	5.127	0.001

the displayed precision. The agreement between the two implementations is expected because both fit the same ordinary least squares multiple regression model using the same data and predictors.

For Auto MPG, the full model explained approximately 82.1% of the variation in MPG. Weight had the largest absolute correlation with MPG among the predictors and produced a stronger simple regression fit than displacement. For Concrete, the full model explained approximately 61.5% of the variation in compressive strength, while the individual simple regression models explained substantially less variation. For Airfoil, the full model explained approximately 51.6% of the variation in scaled sound pressure, while the individual simple regression models explained less than 16%.

The results also demonstrate the importance of considering relationships among predictors rather than evaluating coefficients independently. Auto MPG had particularly strong multicollinearity among cylinders, displacement, horsepower, and weight, while Concrete had moderate multicollinearity among several material composition variables. The Airfoil predictors showed comparatively lower multicollinearity. These differences help explain why the full regression models captured more variation in the response than the individual simple regressions, while multicollinearity made some individual coefficient estimates more difficult to interpret.

3 Regularized Regression

Regularized regression is used to reduce overfitting by adding a penalty to the regression coefficients. For the Auto MPG dataset, Ridge and Lasso regression are applied to examine how coefficient shrinkage affects model performance when predicting MPG. Ridge shrinks predictor coefficients toward zero, while Lasso can additionally reduce coefficients to exactly zero, effectively performing variable selection. Both methods are implemented using `ScalaTion` and `statsmodels`.

Table 6: Concrete full regression coefficients from **statsmodels**.

Predictor	Estimate	Std. Error	t	p
Intercept	-23.1638	26.588	-0.871	0.384
Cement	0.1198	0.008	14.110	0.001
Blast furnace slag	0.1038	0.010	10.245	0.001
Fly ash	0.0879	0.013	6.988	0.001
Water	-0.1503	0.040	-3.741	0.001
Superplasticizer	0.2907	0.093	3.110	0.002
Coarse aggregate	0.0180	0.009	1.919	0.055
Fine aggregate	0.0202	0.011	1.883	0.060
Age	0.1142	0.005	21.046	0.001

Table 7: Airfoil full regression coefficients from **statsmodels**.

Predictor	Estimate	Std. Error	t	p
Intercept	132.8338	0.545	243.866	0.001
Frequency	-0.0013	0.00004	-30.452	0.001
Angle of attack	-0.4219	0.039	-10.847	0.001
Chord length	-35.6880	1.630	-21.889	0.001
Free-stream velocity	0.0999	0.008	12.279	0.001
Suction displacement thickness	-147.3005	15.015	-9.810	0.001

3.1 Statsmodels

3.1.1 Ridge Regression

Alpha: 0.001, Average RMSE: 3.3532
Alpha: 0.01, Average RMSE: 3.3490
Alpha: 0.1, Average RMSE: 3.4184
Alpha: 1, Average RMSE: 3.8973
Alpha: 10, Average RMSE: 6.1356
Alpha: 100, Average RMSE: 7.6559

Best Ridge Alpha: 0.01

Figure 7: Ridge alpha tuning using 5-fold cross-validation.

The Ridge regularization parameter was tuned using 5-fold cross-validation on the training data. Candidate values of $\alpha \in \{0.001, 0.01, 0.1, 1, 10, 100\}$ were evaluated using average validation RMSE. The value $\alpha = 0.01$ produced the lowest average RMSE and was selected for the final Ridge model.

```

const          23.599361
cylinders      -0.469692
displacement   0.977049
horsepower     -0.908489
weight         -4.667132
acceleration   0.006313
model_year     2.722611
origin         1.268601
dtype: float64

```

Figure 8: Ridge regression coefficients.

The fitted Ridge model retains all predictors while shrinking their coefficients through regularization. Among the standardized predictors, weight has the largest coefficient magnitude at approximately -4.67 , indicating a strong negative association with MPG, while model year has a relatively large positive coefficient of approximately 2.72 . Acceleration has a coefficient close to zero, indicating a relatively small contribution to the fitted model.

	Actual MPG	Predicted MPG
0	26.0	25.890771
1	21.6	26.167406
2	36.1	34.297227
3	26.0	24.800514
4	27.0	28.494060

Figure 9: Actual and predicted MPG values from the Ridge model.

The fitted output compares the actual MPG values from the test data with the MPG values predicted by the Ridge model. The predictions generally follow the observed MPG values, although some prediction error remains.

```

Ridge RMSE: 3.2973096237408797
Ridge R²: 0.7869881168144173

```

Figure 10: Ridge regression evaluation metrics.

Ridge regression achieved an R^2 of 0.787, meaning the model explains approximately 78.7% of the variation in MPG in the test data. The RMSE of 3.297 indicates that the model's MPG predictions have a typical error magnitude of about 3.3 MPG.

Overall, Ridge regression provides a reasonably strong fit while using regularization to shrink the predictor coefficients and reduce the risk of overfitting. This is particularly useful for the Auto MPG data because several predictors, including weight, displacement, horsepower, and cylinders, are strongly related to one another, as shown in Project 1. Ridge allows the model to retain all of these predictors while shrinking their coefficients rather than eliminating them. Thus, Ridge maintains information from the different vehicle characteristics while controlling the influence of highly related predictors.

3.1.2 Lasso Regression

```
Alpha: 0.001, Average RMSE: 3.3536
Alpha: 0.01, Average RMSE: 3.3437
Alpha: 0.1, Average RMSE: 3.3518
Alpha: 1, Average RMSE: 3.6104
Alpha: 10, Average RMSE: 7.9141
Alpha: 100, Average RMSE: 7.9141

Best Lasso Alpha: 0.01
```

Figure 11: Lasso alpha tuning using 5-fold cross-validation.

The Lasso regularization parameter was tuned using 5-fold cross-validation on the training data. Candidate values of $\alpha \in \{0.001, 0.01, 0.1, 1, 10, 100\}$ were evaluated using average validation RMSE. The value $\alpha = 0.01$ produced the lowest average RMSE and was selected for the final Lasso model.

```
const          23.599361
cylinders      -0.382769
displacement   1.128755
horsepower     -0.839027
weight        -4.975586
acceleration   0.000000
model_year     2.764608
origin         1.258881
dtype: float64
```

Figure 12: Lasso regression coefficients.

The fitted Lasso model shrinks the predictor coefficients through regularization and reduces the coefficient for acceleration to zero, effectively removing it as a predictor. Weight has the largest coefficient magnitude among the remaining standardized predictors, while model year has a relatively large positive coefficient.

	Actual MPG	Predicted MPG
0	26.0	25.837619
1	21.6	26.086561
2	36.1	34.424332
3	26.0	24.822670
4	27.0	28.448806

Figure 13: Actual and predicted MPG values from the Lasso model.

The fitted output compares the actual MPG values from the test data with the MPG values predicted by the Lasso model. The predicted values generally follow the actual MPG values while still showing some prediction error.

Lasso RMSE: 3.2906920560669057
Lasso R^2 : 0.7878422713750637

Figure 14: Lasso regression evaluation metrics.

Lasso regression achieved an R^2 of 0.7878, meaning the model explains approximately 78.8% of the variation in MPG in the test data. The RMSE of 3.291 indicates that the model's MPG predictions have a typical error magnitude of about 3.29 MPG.

Overall, Lasso regression provides a reasonably strong fit while using regularization to shrink coefficients. In this model, Lasso reduced the coefficient for acceleration to zero, effectively removing it as a predictor. This is consistent with the Auto MPG analysis in Project 1, where acceleration had a weaker relationship with MPG than predictors such as weight, displacement, horsepower, and cylinders. Thus, Lasso maintained strong predictive performance while producing a more selective model.

3.2 ScalaTion

3.2.1 Ridge Regression

```
Ridge lambda tuning:
| RidgeRegression with lambda = 0.1 has sse = 713.7525699365118 |
| RidgeRegression with lambda = 0.2 has sse = 713.5365867814872 |
| RidgeRegression with lambda = 0.4 has sse = 713.1495239145677 |
| RidgeRegression with lambda = 0.8 has sse = 712.5362970863046 |
| RidgeRegression with lambda = 1.6 has sse = 711.8303659095648 |
| RidgeRegression with lambda = 3.2 has sse = 711.8153255080763 |
| RidgeRegression with lambda = 6.4 has sse = 714.4740228233744 |
| RidgeRegression with lambda = 12.8 has sse = 722.4991830313115 |
| RidgeRegression with lambda = 25.6 has sse = 737.7183111025892 |
| RidgeRegression with lambda = 51.2 has sse = 762.7016504571261 |
| RidgeRegression with lambda = 102.4 has sse = 807.6552158726756 |
| RidgeRegression with lambda = 204.8 has sse = 897.1513975453229 |
| RidgeRegression with lambda = 409.6 has sse = 1076.40580565289 |
| RidgeRegression with lambda = 819.2 has sse = 1409.7306277835444 |
| RidgeRegression with lambda = 1638.4 has sse = 1919.7235723580118 |
| RidgeRegression with lambda = 3276.8 has sse = 2505.485491122153 |
| RidgeRegression with lambda = 6553.6 has sse = 3010.629751886936 |
| RidgeRegression with lambda = 13107.2 has sse = 3359.4603911741638 |
| RidgeRegression with lambda = 26214.4 has sse = 3567.8769296775936 |
| RidgeRegression with lambda = 52428.8 has sse = 3682.349877269195 |

Best Ridge Lambda: 3.2
Best Ridge CV SSE: 711.8153
```

Figure 15: ScalaTion Ridge lambda tuning using cross-validation.

The ScalaTion Ridge regularization parameter was tuned using cross-validation on the training data. Candidate lambda values were evaluated using cross-validated SSE, where lower SSE indicates better validation performance. The lowest CV SSE of 711.815 was obtained at $\lambda = 3.2$, which was selected for the final Ridge model.

```
Ridge coefficients:
Intercept = 23.416561
cylinders      -1.078994
displacement   1.232053
horsepower     -0.542461
weight        -4.738602
acceleration   0.007586
model_year     2.651239
origin         1.139237

Ridge actual vs predicted MPG:
Actual: 34.300 Predicted: 31.613
Actual: 19.000 Predicted: 26.525
Actual: 15.000 Predicted: 15.493
Actual: 33.700 Predicted: 33.805
Actual: 29.800 Predicted: 33.607

Ridge RMSE: 3.556045
Ridge R²: 0.779972
```

Figure 16: ScalaTion Ridge regression coefficients, fitted output, and evaluation metrics.

The ScalaTion Ridge model achieved an R^2 of 0.7800, meaning the model explains approximately 78.0% of the variation in MPG in the test data. The RMSE of 3.556 indicates that the model's MPG predictions have a typical error magnitude of about 3.56 MPG. The fitted output also shows that the predicted MPG values generally follow the actual test values, although some prediction error remains.

Overall, the ScalaTion Ridge model provides a reasonably strong fit while retaining all predictors and shrinking their coefficients through regularization. Weight has the largest coefficient magnitude among the standardized predictors at approximately -4.739 , indicating a strong negative association with MPG in the fitted model. Model year has a positive coefficient of approximately 2.651, while acceleration has a coefficient close to zero. As with the Statsmodels Ridge model, Ridge retains these predictors rather than eliminating them.

3.2.2 Lasso Regression

```
Lasso lambda tuning:
| LassoRegression with lambda = 3.2 has sse = 716.4706211833529 |
| LassoRegression with lambda = 6.4 has sse = 715.9913823683123 |
| LassoRegression with lambda = 12.8 has sse = 716.2214580015408 |
| LassoRegression with lambda = 25.6 has sse = 715.414402699883 |
| LassoRegression with lambda = 51.2 has sse = 718.6457373741941 |
| LassoRegression with lambda = 102.4 has sse = 739.1994583170452 |
| LassoRegression with lambda = 204.8 has sse = 827.8038274144692 |
| LassoRegression with lambda = 409.6 has sse = 1147.4621449313456 |
| LassoRegression with lambda = 819.2 has sse = 2378.1516869310367 |
| LassoRegression with lambda = 1638.4 has sse = 6140.51301731296 |
| LassoRegression with lambda = 3276.8 has sse = 14132.35192204406 |
| LassoRegression with lambda = 6553.6 has sse = 37826.29469091059 |
| LassoRegression with lambda = 13107.2 has sse = 37826.29469091059 |
| LassoRegression with lambda = 26214.4 has sse = 37826.29469091059 |
| LassoRegression with lambda = 52428.8 has sse = 37826.29469091059 |
| LassoRegression with lambda = 104857.6 has sse = 37826.29469091059 |
| LassoRegression with lambda = 209715.2 has sse = 37826.29469091059 |
| LassoRegression with lambda = 419430.4 has sse = 37826.29469091059 |
| LassoRegression with lambda = 838860.8 has sse = 37826.29469091059 |
| LassoRegression with lambda = 1677721.6 has sse = 37826.29469091059 |

Best Lasso Lambda: 25.6
Best Lasso CV SSE: 715.4144
```

Figure 17: ScalaTion Lasso lambda tuning using cross-validation.

The ScalaTion Lasso regularization parameter was tuned using cross-validation on the training data. Candidate lambda values were evaluated using cross-validated SSE, where lower SSE indicates better validation performance. The lowest CV SSE of 715.414 was obtained at $\lambda = 25.6$, which was selected for the final Lasso model.

```
Lasso coefficients:
const          23.335032
cylinders      -0.250666
displacement   -0.143925
horsepower     -0.121958
weight        -4.643400
acceleration    0.050525
model_year     2.601568
origin         0.943053

Lasso actual vs predicted MPG:
Actual: 34.300 Predicted: 31.550
Actual: 19.000 Predicted: 26.369
Actual: 15.000 Predicted: 15.727
Actual: 33.700 Predicted: 33.301
Actual: 29.800 Predicted: 33.434

Lasso RMSE: 3.544011
Lasso R2: 0.781459
```

Figure 18: ScalaTion Lasso regression coefficients, fitted output, and evaluation metrics.

The ScalaTion Lasso model achieved an R^2 of 0.7815, meaning the model explains approximately 78.1% of the variation in MPG in the test data. The RMSE of 3.544 indicates that the model's MPG predictions have a typical error magnitude of about 3.54 MPG. The fitted output similarly shows that the predicted MPG values generally follow the actual test values.

Overall, the ScalaTion Lasso model provides a reasonably strong fit while shrinking the predictor coefficients through regularization. Weight has the largest coefficient magnitude among the standardized predictors at approximately -4.643 , while model year has a positive coefficient of approximately 2.602 . Unlike the Statsmodels Lasso model, the ScalaTion Lasso model did not reduce acceleration exactly to zero. Its coefficient remained small at approximately 0.051 . The ScalaTion Ridge and Lasso models therefore produced very similar predictive performance, although the two regularization methods produced different coefficient estimates.

3.3 Summary

Overall, `statsmodels` and ScalaTion produced similar Ridge and Lasso results for the Auto MPG dataset. The `statsmodels` models had slightly stronger test performance. Ridge achieved an RMSE of 3.297 and R^2 of 0.7870 in `statsmodels`, compared with an RMSE of 3.556 and R^2 of 0.7800 in ScalaTion. Lasso achieved an RMSE of 3.291 and R^2 of 0.7878 in `statsmodels`, compared with an RMSE of 3.544 and R^2 of 0.7815 in ScalaTion. Although the results are close, the `statsmodels` models produced slightly lower prediction error and explained slightly more variation in MPG in these runs.

The two implementations also used different regularization parameters and tuning procedures. In `statsmodels`, the penalty strength is controlled by α , and was selected using 5-fold cross-validation with average validation RMSE. In ScalaTion, the penalty strength is controlled by λ , and the built-in tuning procedure selects λ using cross-validated SSE. Because the implementations and tuning criteria differ, the numerical values of α and λ should not be directly compared.

Ridge behaved similarly across both implementations by shrinking coefficients while retaining all predictors. Lasso showed a larger difference: the `statsmodels` model reduced acceleration to exactly zero, while ScalaTion retained acceleration with a small coefficient of approximately 0.051 . Within both implementations, Ridge and Lasso had very similar predictive performance, with Lasso having a slightly lower RMSE and slightly higher R^2 . Overall, the results show that both methods provide similar predictive performance, while Lasso can produce a more selective model depending on the implementation and selected regularization strength.

4 Forward Feature Selection

TODO

5 Transformed Regression

TODO

6 Symbolic Regression

Symbolic regression differs from traditional linear regression because it searches not only for model coefficients, but also for the mathematical structure relating the predictors to the response. Instead of assuming a predefined functional form, symbolic regression searches over combinations of mathematical operators and variables to identify equations that balance predictive accuracy and expression complexity.

Two symbolic regression implementations were used in this project: ScalaTion and PySR. ScalaTion constructs transformed candidate features and applies feature selection, while PySR uses an evolutionary search procedure to discover mathematical expressions directly. The three datasets analyzed were Auto MPG, Concrete Compressive Strength, and Airfoil Self-Noise.

6.1 ScalaTion

ScalaTion symbolic regression was applied to all three datasets. Candidate features were constructed from nonlinear transformations of the original predictors, including square-root, squared, inverse, logarithmic, and interaction terms depending on the dataset. Forward feature selection was then used to identify a subset of transformed predictors for the final regression model.

Table 8: ScalaTion symbolic regression performance.

Dataset	R^2	Adjusted R^2	RMSE	MAE
Auto MPG	0.8827	0.8328	2.6101	1.9914
Concrete	0.8180	0.8015	6.9646	5.6196
Airfoil	0.6826	0.6321	3.9647	2.9289

6.1.1 Auto MPG

For Auto MPG, ScalaTion generated nonlinear transformations of vehicle characteristics including weight, model year, displacement, horsepower, acceleration, cylinders, and origin. The final selected model included inverse, square-root, and squared transformations.

Table 9: Selected significant coefficients from the ScalaTion symbolic regression model for Auto MPG.

Term	Estimate	Std. Error	t	p
Intercept	-1016807.1951	517225.1602	-1.97	0.0493
Weight ⁻¹	908334.5295	426892.0794	2.13	0.0334
Model year ²	-10.1170	4.9912	-2.03	0.0427
$\sqrt{\text{model year}}$	83706.6823	42238.6395	1.98	0.0475
Acceleration ⁻¹	808.9958	331.6133	2.44	0.0147
Displacement ⁻²	129732.1150	56783.3200	2.28	0.0223
Weight ⁻²	-626524289.4326	262993286.5486	-2.38	0.0172
Acceleration ⁻²	-3554.8823	1663.9678	-2.14	0.0326
Acceleration ²	0.02863	0.01384	2.07	0.0386
Cylinders ²	1.1834	0.2464	4.80	< 0.001
Cylinders ⁻²	-765.5316	127.7204	-5.99	< 0.001
$\sqrt{\text{cylinders}}$	-114.1268	21.5207	-5.30	< 0.001

The final selected model achieved an R^2 of 0.8827 and an adjusted R^2 of 0.8328. The RMSE was 2.6101 MPG and the MAE was 1.9914 MPG. The residual standard error was approximately 3.1370, and the overall F -statistic was 17.6756.

Several transformed predictors were statistically significant, including inverse weight, squared model year, inverse acceleration, inverse displacement, squared acceleration, and multiple nonlinear transformations of cylinders. These results indicate that the relationship between vehicle characteristics and MPG contains important nonlinear structure.

However, many of the generated transformed terms had extremely large VIF values, with several exceeding millions or even billions. This indicates severe multicollinearity among the symbolic basis functions. Therefore, individual coefficient magnitudes should be interpreted cautiously, and the model is better viewed as a predictive combination of nonlinear transformed features.

6.1.2 Concrete

For the Concrete dataset, ScalaTion generated transformations of cement, slag, fly ash, water, superplasticizer, coarse aggregate, fine aggregate, and age. The selected model contained terms such as $\sqrt{\text{age}}$, cement squared, slag squared, fly ash squared, square-root transformations, and other transformed material variables.

Table 10: Selected significant coefficients from the ScalaTion symbolic regression model for Concrete.

Term	Estimate	Std. Error	t	p
$\sqrt{\text{age}}$	6.6596	0.2074	32.11	< 0.001
Superplasticizer	-1.6063	0.4672	-3.44	< 0.001
Cement ²	0.000105	0.000007	14.14	< 0.001
Age	-0.2441	0.0119	-20.52	< 0.001
Slag ²	0.000142	0.000022	6.30	< 0.001
$\sqrt{\text{water}}$	564.3547	79.3692	7.11	< 0.001
$\sqrt{\text{fine aggregate}}$	-2.1050	0.3517	-5.99	< 0.001
$\sqrt{\text{superplasticizer}}$	6.5834	1.2153	5.42	< 0.001
Water	-32.4357	4.4639	-7.27	< 0.001
Water ²	0.03036	0.00414	7.34	< 0.001

The final selected model achieved an R^2 of 0.8180 and an adjusted R^2 of 0.8015. The RMSE was 6.9646 MPa and the MAE was 5.6196 MPa. The residual standard error was approximately 7.2904, and the overall F -statistic was 49.6891.

For comparison, the full 24-term candidate model produced a slightly higher test R^2 of 0.8378, with an RMSE of 6.4876 and an MAE of 5.0248. However, the reported symbolic regression results focus on the selected feature-subset model, which represents the final model retained by the ScalaTion selection procedure.

Several nonlinear terms were statistically significant, including $\sqrt{\text{age}}$, cement², slag², $\sqrt{\text{water}}$, and water².

6.1.3 Airfoil

The Airfoil symbolic regression model included transformed versions and interactions among frequency, angle of attack, chord length, free-stream velocity, and displacement thickness. Candidate terms included logarithmic, square-root, squared, and multi-variable interaction terms.

Table 11: Selected significant coefficients from the ScalaTion symbolic regression model for Airfoil.

Term	Estimate	Std. Error	t	p
Displacement thickness $\times$ chord length $\times$ frequency	0.7788	0.1640	4.75	< 0.001
Chord length $\times$ frequency	-0.0127	0.0016	-7.81	< 0.001
Attack angle ²	0.1499	0.0302	4.96	< 0.001
Velocity $\times$ chord length $\times$ frequency	0.000171	0.000027	6.34	< 0.001
Custom transformed term	-2.2738	0.2685	-8.47	< 0.001
Chord length $\times$ attack angle $\times$ frequency	-0.00159	0.00020	-8.09	< 0.001
Displacement thickness $\times$ velocity $\times$ attack angle	-0.6896	0.2485	-2.78	0.0055
Displacement thickness $\times$ frequency	-0.1888	0.0456	-4.14	< 0.001
Displacement thickness $\times$ attack angle $\times$ frequency	0.00714	0.00192	3.71	< 0.001
Velocity $\times$ chord length $\times$ attack angle	0.1275	0.0352	3.62	< 0.001
Attack angle ^{1/2}	-24.9059	7.6452	-3.26	0.0011
Frequency	0.00577	0.00142	4.07	< 0.001
Velocity $\times$ chord length	-0.7778	0.1716	-4.53	< 0.001
Velocity ^{1/2}	-3.5519	0.8407	-4.22	< 0.001
Attack angle	280.2162	124.8255	2.24	0.0248
Displacement thickness ^{1/2}	35.0415	10.3336	3.39	< 0.001
Displacement thickness $\times$ velocity	18.7152	5.5923	3.35	< 0.001
Velocity $\times$ attack angle	-0.01508	0.00450	-3.35	< 0.001
Displacement thickness $\times$ velocity $\times$ frequency	-0.000908	0.000452	-2.01	0.0445
Displacement thickness $\times$ chord length	-6170.3137	2631.6606	-2.34	0.0190
$\log(1 + \text{frequency})$	8.6614	1.8632	4.65	< 0.001
$\sqrt{\text{frequency}}$	-0.9257	0.2086	-4.44	< 0.001

The final model achieved an R^2 of 0.6826 and an adjusted R^2 of 0.6321. The RMSE was 3.9647 dB and the MAE was 2.9289 dB. The residual standard error was approximately 4.2752 and the overall F -statistic

was 13.5314.

Although the Airfoil model explained less variation than the Auto MPG and Concrete ScalaTion models, it still captured substantial nonlinear structure in the response.

Many of the selected transformed features were statistically significant, indicating that the Airfoil response contains substantial nonlinear and interaction effects. However, several transformed terms had extremely large VIF values, with some approaching infinity. This indicates severe multicollinearity among the generated symbolic features. Therefore, the individual coefficient magnitudes should not be interpreted independently; the model is better viewed as a predictive combination of nonlinear basis functions.

6.2 PySR

PySR was also used to perform symbolic regression on all three datasets. Each dataset was divided into training and testing sets using an 80/20 split with `random_state=42`. PySR searches for mathematical expressions through an evolutionary procedure and evaluates candidate expressions based on predictive loss and expression complexity.

The operator set included the binary operators

$$+, -, \times, /$$

and nonlinear unary transformations including square root and squared terms. For the Concrete and Airfoil datasets, logarithmic transformations were also included.

6.2.1 Auto MPG

The Auto MPG dataset contained 392 observations and seven predictors: displacement, cylinders, horsepower, weight, acceleration, model year, and origin. The data were divided into 313 training observations and 79 testing observations.

PySR was run for 150 iterations with the operators `+`, `-`, `*`, and `/`, together with the `square` and `sqrt` unary operators. Expression complexity was limited using `maxsize=30`, and model selection was performed using the `best` criterion.

The selected equation was

$$\widehat{\text{MPG}} = \frac{2149.9915(\text{model_year} - 46.281776)}{\text{weight}}. \quad (1)$$

This expression indicates that predicted fuel economy increases with model year and decreases with vehicle weight. Despite the availability of seven predictors, PySR selected a compact equation involving only model year and weight.

On the held-out test set, the model achieved

$$\text{MSE} = 7.5596, \quad \text{RMSE} = 2.7495,$$

$$\text{MAE} = 2.0550, \quad R^2 = 0.8519.$$

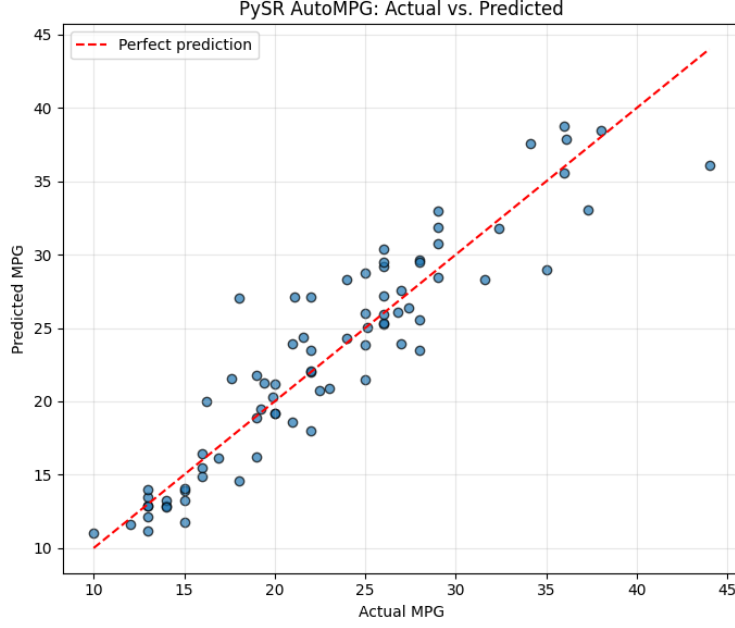


Figure 19: Actual versus predicted MPG values for the PySR symbolic regression model. The dashed diagonal line represents perfect prediction.

The predicted MPG values generally follow the perfect-prediction line, indicating that the compact two-variable equation captures much of the variation in MPG.

6.2.2 Concrete

The Concrete dataset contained 1030 observations and eight predictors: cement, blast furnace slag, fly ash, water, superplasticizer, coarse aggregate, fine aggregate, and age. The dataset was divided into 824 training observations and 206 testing observations.

PySR was run for 300 iterations with arithmetic operators and the nonlinear operators `square`, `sqrt`, and `log`. The final equation selected using the lowest training loss was

$$\widehat{\text{Strength}} = -0.200393 \text{ water} + \sqrt{\text{age superplasticizer}^{1/16}} \quad (2)$$

$$+ 1.201766 \sqrt{\sqrt{\text{fly_ash}} + \text{slag} - \frac{2.197127 \text{ slag}}{\text{age}}} \quad (3)$$

$$+ 6.627109 \log(\text{age}) \quad (4)$$

$$+ 1.201766 \log^2(\text{cement}) \log^2 \left(\log \left(\sqrt{\text{cement}} \right) \right). \quad (5)$$

The expression captures several nonlinear relationships. Water contributes negatively, while age enters through both square-root and logarithmic terms. The model also includes nonlinear contributions from slag, fly ash, superplasticizer, and cement.

The held-out test performance was

$$\text{MSE} = 42.3301, \quad \text{RMSE} = 6.5062,$$

$$\text{MAE} = 5.3603, \quad R^2 = 0.8357.$$

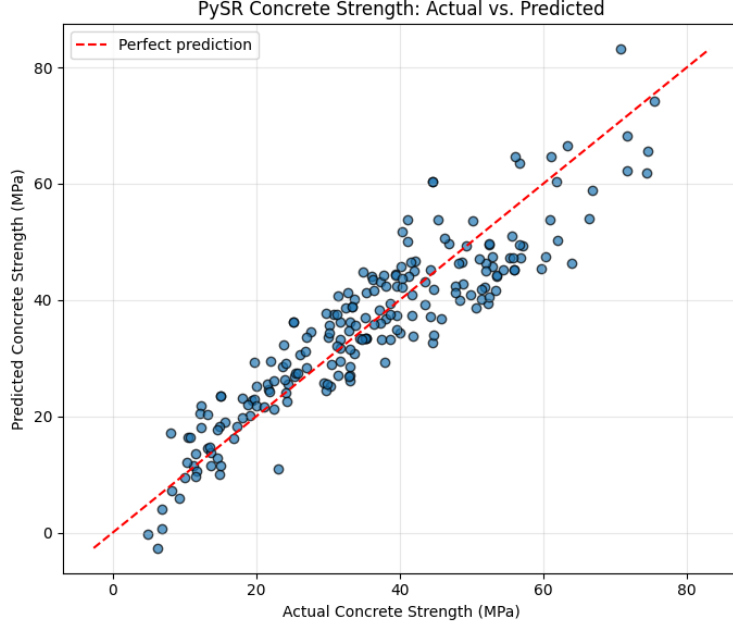


Figure 20: Actual versus predicted concrete compressive strength from the PySR symbolic regression model.

The model explains approximately 83.6% of the variation in compressive strength on the test data, indicating that the nonlinear expression captures important relationships that are not represented by a simple linear model.

6.2.3 Airfoil

The Airfoil Self-Noise dataset contained 1503 observations with five predictors: frequency, angle of attack, chord length, free-stream velocity, and displacement thickness. The data were divided into 1202 training observations and 301 testing observations.

PySR was run for 500 iterations using arithmetic operators together with square, square-root, and logarithmic transformations. The selected symbolic expression was

$$\widehat{\text{SPL}} = - \frac{36.989017 f}{f + (-195.3614 d^2 v + v)^2 + 128.70335 + \frac{2.3002949}{cd}} \quad (6)$$

$$+ 136.76428 - \frac{530.60527}{cf \log(195.3614d/c)}, \quad (7)$$

where f represents frequency, v represents velocity, c represents chord length, and d represents displacement thickness.

The model obtained the following test performance:

$$\text{MSE} = 9.2244, \quad \text{RMSE} = 3.0372,$$

$$\text{MAE} = 2.3529, \quad R^2 = 0.8159.$$

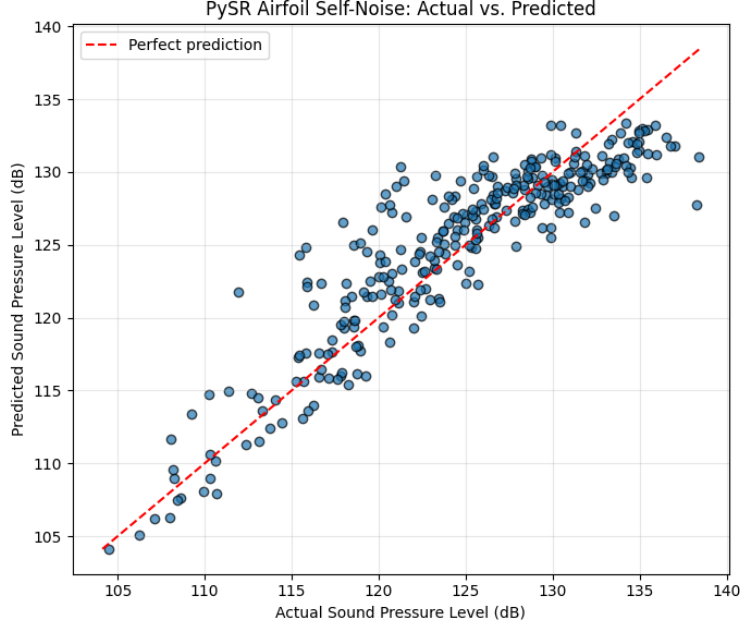


Figure 21: Actual versus predicted sound pressure levels from the PySR symbolic regression model.

The symbolic equation combines frequency, velocity, chord length, and displacement thickness through nonlinear ratios and logarithmic terms. The model explains approximately 81.6% of the variation in sound pressure level on the test set.

6.3 Comparison of ScalaTion and PySR

Table 12: Comparison of ScalaTion and PySR symbolic regression performance.

Dataset	Method	R^2	RMSE
Auto MPG	ScalaTion	0.8827	2.6101
	PySR	0.8519	2.7495
Concrete	ScalaTion	0.8180	6.9646
	PySR	0.8357	6.5062
Airfoil	ScalaTion	0.6826	3.9647
	PySR	0.8159	3.0372

The two symbolic regression implementations produced different equation structures because they use different search procedures. ScalaTion constructs a large transformed feature matrix and performs feature selection, whereas PySR directly searches over mathematical expressions.

For Auto MPG, both methods produced strong predictive performance. ScalaTion reported the larger R^2 , while PySR produced a substantially more compact equation involving only model year and weight. For Concrete, PySR achieved a higher R^2 and lower RMSE than the ScalaTion selected model. The largest difference occurred for the Airfoil dataset, where PySR achieved an R^2 of 0.8159 compared with 0.6826 for ScalaTion.

Overall, the results demonstrate one of the major advantages of symbolic regression: the ability to discover nonlinear functional relationships while still producing explicit mathematical expressions that can be inspected and interpreted.